Between Buses: A Toilet Within Reach?

Public toilet accessibility at frequent bus transfer locations in Hong Kong

Where do frequent bus transfers coincide with poor toilet access?

How does this relationship change throughout the day?

Unit of observation

A passenger transfer event linked to a location and time.
Events are aggregated into consistent observation intervals.

Measurement focus

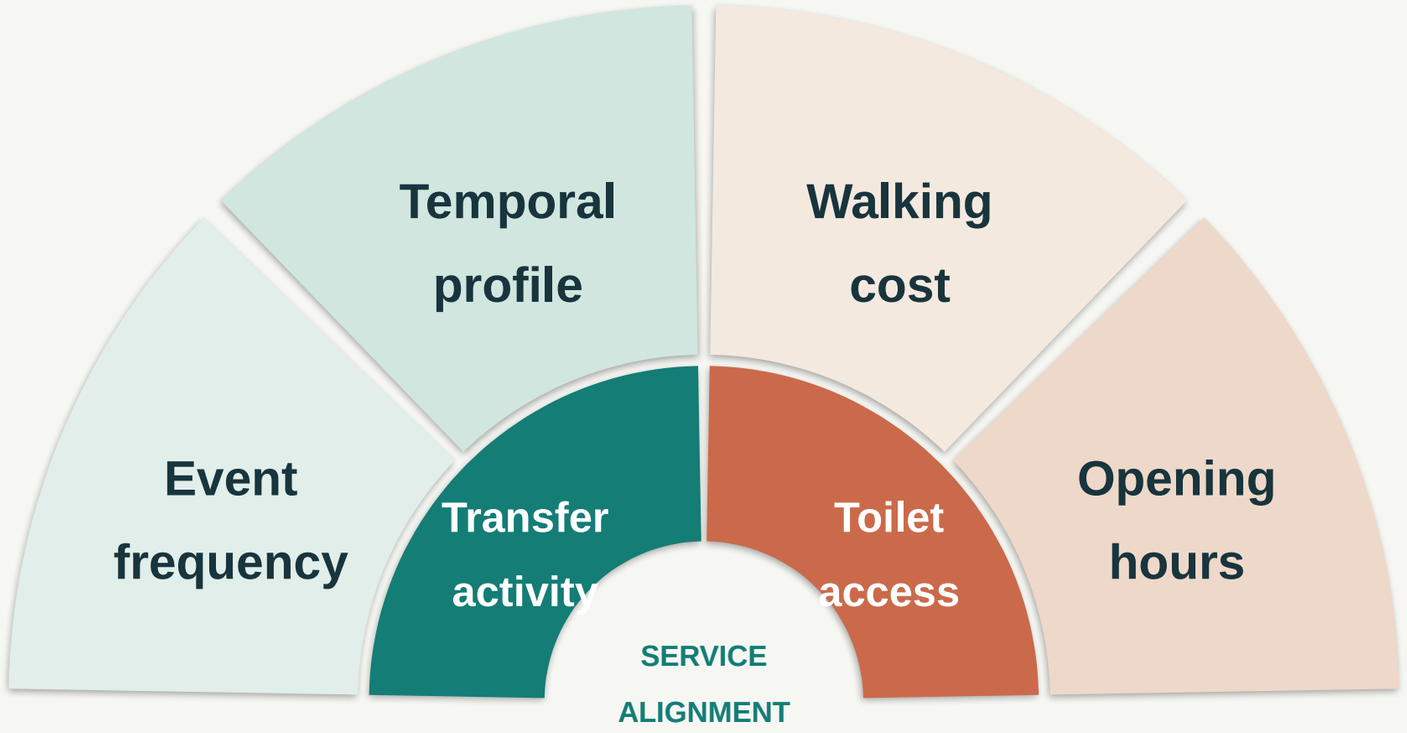
Transfer frequency and access to open public toilets
within the pedestrian network.

Activity meets accessibility

Transfer events describe where and when passengers change services. Accessibility describes the available toilet service.

When transfers concentrate

Where transfers occur frequently



Effort to reach a toilet

Facilities open at that time

The intersection identifies busy transfer locations with limited toilet access.

Conceptual framework. Sector size carries no numerical meaning.

Where and when do passengers transfer?

Frequent transfer locations are identified from passenger events within comparable observation periods.

Event

A valid connection

Link successive journey legs to a transfer location and time.

Location

Stops and interchange areas

Aggregate nearby stops while retaining boarding direction.

Time

Consistent intervals

Compare hourly activity and recurring daily patterns.

Measures

Transfer counts within each interval

Peak concentration and weekday patterns

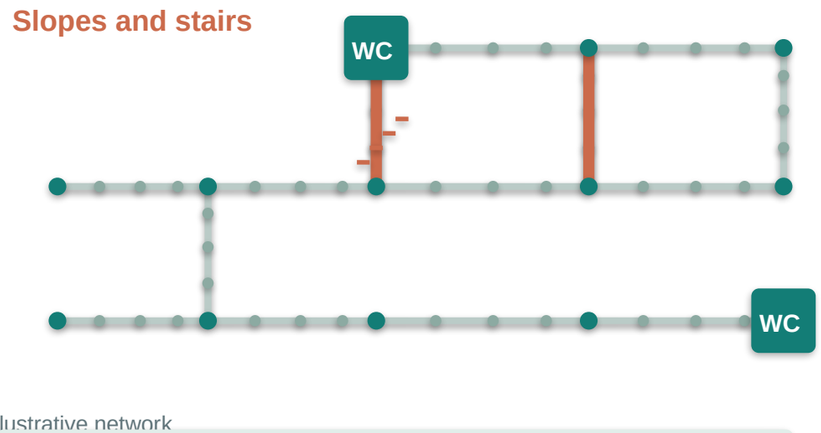
Persistence of high frequency locations across days

Passenger transfer frequency provides the activity weight used in the accessibility assessment.

How far is a toilet from the transfer?

Access is measured from the actual transfer location to a public toilet that is open at the event time.

Access through the pedestrian network



Illustrative network

Stop position → pedestrian route → toilet entrance

Measures

Distance to the nearest open toilet

Walking time along a feasible route

Access within selected time thresholds

Spatial conditions

Slopes and stairs

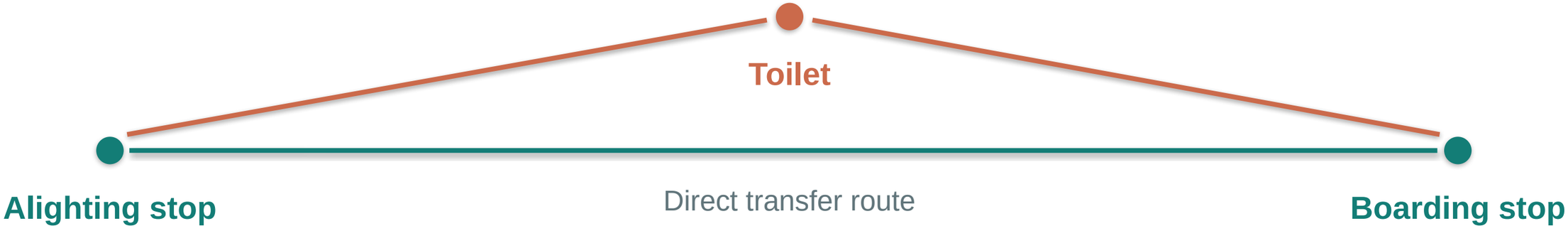
Road crossings and vertical connections

Stop groups summarize activity. Specific stop positions determine walking access.

Sun et al. (2021)

What does a toilet visit add to the transfer?

Paired alighting and boarding locations allow access to be assessed within the transfer journey.



Measures

Additional walking distance or time via a toilet

Differences between transfer directions

Variation with toilet opening hours

With aggregate stop counts only, the assessment uses access from the transfer location.

Is toilet access available when transfers peak?

Transfer activity and toilet availability are aligned within the same time intervals.

Measure	Interpretation	Comparison
Event coverage	Share of transfers within the access threshold	Across hours
Uncovered events	Transfers at locations beyond the threshold	Across locations
Peak service gap	Access during periods of concentrated transfers	Peak and off peak
Opening alignment	Access gained or lost as toilets open or close	Across day types

Counts express the scale of the access gap. Proportions express its prevalence among transfers.

Which changes serve the most transfers?

Alternative service arrangements are compared against the same observed transfer activity.

Location

Additional toilets near frequent transfer locations

Opening hours

Extended availability during periods of concentrated activity

Pedestrian access

Improved entrances or links between stops and toilets

Measures

Additional transfer events within the access threshold

Reduction in walking cost at frequent transfer locations

Distribution of gains across locations and time periods

The assessment prioritizes service gaps using both transfer volume and accessibility.

Bruno et al. (2024)